

Class 28 Functions

Goals

1. Know the definition of function and inverse function.
2. Know what it means for a function to be one-to-one and onto.
3. Know what it means for two functions to be equal.
4. Be able to determine if a given relation is a function, is one-to-one, is onto.

Definitions and Terminology

Definition: Let D and Y be nonempty sets. A function f :

$D \rightarrow Y$ is

a nonempty subset of the cartesian product $D \times Y$ in which every $x \in D$ appears in exactly one ordered pair in the subset.

Examples: Let $D = \{1, 2, 3, 4\}$ and $Y = \{2, 4, 6, 8\}$. Make up two examples of functions from D to Y . Make up two examples of mappings from D to Y that are **not** functions.

Definition: If f is a function mapping D to Y with range R , then the inverse function for f is a function g mapping R to D that satisfies

$$(g \circ f)(x) = x \quad \forall x \in D \quad \text{and} \quad (f \circ g)(y) = y \quad \forall y \in R$$

Definition: A function f mapping D to Y is one-to-one means

for all a, b in D $f(a) = f(b)$ implies $a = b$.
Equivalently if $a \neq b$, then $f(a) \neq f(b)$.

A one-to-one function is also called an injection.

Definition: $f: D \rightarrow Y$ is onto means

For all $y \in Y$, there is some x in D such that $y = f(x)$
So every y in Y shows up in at least one ordered pair.

An onto function is also called an surjection.

A function that is both one-to-one and onto is called a bijection.

Definition: Functions f and g are equal means

They have the same domain and $f(x) = g(x)$ for every x in the domain.

Examples: Decide if each function is one-to-one, onto, both, or neither.

$f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^2$

neither one-to-one nor onto.

$g: \mathbb{R}^+ \rightarrow \mathbb{R}$ defined by $g(x) = \ln(x)$ bijection. $\ln(a) = \ln(b) \Rightarrow e^{\ln(a)} = e^{\ln(b)} \Rightarrow a = b$.

$f: \mathbb{C} \rightarrow \mathbb{R}^2$ defined by $f(a + bi) = a^2 + b^2$.

not one-to-one, but is onto.

$g: M_{2,2} \rightarrow \mathbb{R}$ defined by $g\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = ad - bc$

not one-to-one, but is onto

Practice: prove the following fact about onto functions.

If f and g are onto, then $g \circ f$ is onto.

Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be onto functions and let z be any element of Z .

Since g maps Y onto Z , there is some y in Y such that $g(y) = z$.

Since f maps X onto Y , there is some x in X such that $f(x) = y$.

Therefore $(g \circ f)(x) = g(f(x)) = g(y) = z$.

So $g \circ f$ maps X onto Z